Physics-Informed Vision-Based Target Tracking: Dynamic ROI, Adaptive Noise Modeling, and Sensor-Only Evaluation

Önder İhsan Kul, Undergraduate Researcher, Istanbul Technical University

***Abstract*—Visual target tracking in agile robotics is often limited by strict Size, Weight, and Power (SWaP) constraints. While deep learning models excel at feature extraction, their computational latency and bounding-box jitter introduce phase lag that destabilizes high-frequency closed-loop control. To address this, we present a resource-efficient, low-latency 2-DoF visual tracking framework that entirely bypasses deep-learning overhead. Our architecture couples a classical vision stack—utilizing HSV filtering, Hough transforms, and complete-linkage clustering—with a predicting Kalman filter to dynamically generate regions of interest (ROI), aggressively bounding the pixel-processing space. To guarantee stability during high-speed maneuvers, we introduce a physics-informed Adaptive-R covariance mechanism. By mapping 2D pixel error to non-linear 3D depth variance ($e_z \propto Z^2$), this formulation prevents the filter divergence common in static-noise models. Finally, we derive a conservative, sensor-only mathematical precision bound (r_{th}) to evaluate tracking performance without external motion capture data. Evaluations demonstrate robust precision and variance suppression, establishing a rigorous, computationally lightweight alternative for dynamic robotic tracking.**

***Index Terms*—Target Tracking, Object Detection, State Estimation, Adaptive Covariance, Self-Evaluation Metrics, Non-linear Motion, Outlier Rejection, Robust Tracking**

I. Introduction

In the domain of agile autonomous robotics, real-time visual target tracking remains a critical challenge, particularly for platforms operating under strict Size, Weight, and Power (SWaP) constraints. While active depth sensors such as LiDAR provide highly accurate 3D spatial measurements, their integration is often precluded by payload limits and power budgets in micro-robotic and embedded systems. Consequently, monocular vision systems coupled with multi-degree-of-freedom (DoF) pan-tilt actuation have emerged as a lightweight alternative. However, maintaining continuous target lock during high-speed, non-linear maneuvers requires a perception and control pipeline capable of operating with extremely low latency and high mathematical rigor.

Recent advancements in visual tracking have been heavily dominated by deep learning architectures, such as single-shot detectors (e.g., YOLO) and convolutional neural networks [1], [2]. While these models excel at automated feature extraction [3] in complex environments, their inference latency, computational cost [4] and inherent bounding-box jitter frequently destabilize high-frequency closed-loop control systems [5], [6]. For highly dynamic robotic applications, this perception-to-actuation delay introduces severe phase lag, ultimately resulting in filter divergence and target loss. Therefore, there is a critical need for resource-efficient, classical feature-driven perception pipelines that are tightly integrated with robust, physics-informed state estimators.

To address these limitations, this article presents a low-latency, 2-DoF visual tracking framework that completely bypasses heavy deep-learning overhead. By coupling an optimized classical vision stack (utilizing HSV color-space filtering, Hough Circle Transforms, and complete-linkage clustering) with a continuously predicting Kalman Filter, the system dynamically generates regions of interest (ROI) to aggressively bound the pixel-processing space [7]. Furthermore, recognizing that monocular depth estimation uncertainty expands non-linearly with distance [8], the state estimator abandons static measurement covariance in favor of a physics-informed Adaptive-R matrix [9]. Finally, to rigorously validate the system under real-world conditions where external motion capture (ground truth) is unavailable, performance is evaluated using a mathematically derived, worst-case sensor-only precision bound. Self-evaluation metrics can be utilized to evaluate segmentation model performance with high reliability in visual tracking scenarios [10] where ground-truth measurement can be time-consuming and sensitive to human errors [10], [11].

The primary contributions of this work are three-fold:

- **Low-Latency Closed-Loop Perception:** A highly efficient, feature-driven vision pipeline that utilizes state-predicted dynamic ROI generation and complete linkage clustering to reject false-positive artifacts without the computational penalty of deep learning models.
- **Physics-Informed Adaptive State Estimation:** A dynamic measurement covariance scaling mechanism that maps 2D pixel uncertainty to 3D depth variance ($e_z \propto Z^2$), preventing filter divergence during high-speed chaotic maneuvers.
- **Sensor-Only Performance Validation:** The derivation of a conservative, mathematical worst-case precision threshold (r_{th}) that enables rigorous system evaluation completely independent of canonical ground-truth data.

II. Methodology

A. System Architecture & Framework Overview

The primary objective of the proposed framework is to achieve precise target acquisition, continuous visual detection, and real-time tracking of a planar object using a 2-DOF pan-tilt laser platform. The spatial coordinate frames and physical problem configuration are illustrated in Fig. 1.

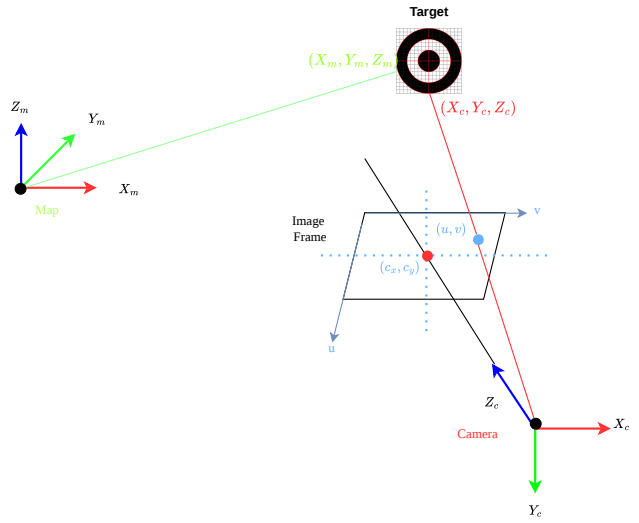


Fig. 1. Spatial coordinate frames and physical tracking problem geometry.

To continuously align the laser tracker with the moving target center throughout dynamic target maneuvers, the system integrates perception, estimation, and control modules into a unified closed-loop pipeline. The end-to-end operational architecture is presented in the block diagram in Fig. 2.

More specifically, in this object tracking problem, the homogenous transformation matrices from map to camera frame $T_{M\to C}$ and camera to laser pointer frame $T_{C\to L}$ are static and known apriori. The system requires accurate estimation of the displacement vector from laser to target frame $t_{L\to T}$ to accurately calculate error angles and subsequently determine angular velocity control commands for pan and tilt axes to consistently track the target under dynamic movement.

To obtain an accurate $t_{L\to T}$ estimate, we construct a Kalman Filter [12], [13] framework where system state is defined as $x_k = [p_x, p_y, p_z, v_x, v_y, v_z]^T$ which represent target position and velocity in the inertial map frame.

The system uses a nearly constant velocity model [14], [15] described by the equations 1-3. ω, G, Q, Q_a represent discrete white noise injected to the system, the noise gain matrix,the process noise covariance matrix and random acceleration noise, respectively.

$$x_k = Fx_{k-1} + G\omega \tag{1}$$

$$F = \begin{bmatrix} 1 & 0 & 0 & dt & 0 & 0 \\ 0 & 1 & 0 & 0 & dt & 0 \\ 0 & 0 & 1 & 0 & 0 & dt \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$Q = GQ_aG^T \tag{3}$$

In this framework, target position measurements are made in the camera frame by solving the PnP problem [16] described in the following section, before measurements are fed to the Kalman Filter, transformed to the map frame using equation 4

$$T_{M\to T} = T_{M\to C}T_{C\to T} \tag{4}$$

After this conversion, H matrix for the filter is specified as in equation 5

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{5}$$

Measurement noise covariance matrix R is adaptive since camera pose estimation uncertainty increases with higher object depth [8], [9]. With σ_r, σ_c representing object detection radius error and object detection center error in pixels respectively, camera measurement errors can be deduced by 1st order approximation on the camera model, explained in equations 6-12. Upper limits for dX_c, dY_c, dZ_c and σ_r, σ_c are defined prior to the simulation based on system assumptions. For our system, max depth estimation error is taken to be 100 cm, corresponding to a $\sigma_{Z_c} = 33$ cm. Similar but much smaller assumptions are made for dX_c and dY_c since errors in these axes are much smaller due to the pinhole camera model [17].

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{p_x}{p_z} \\ \frac{p_y}{p_z} \\ 1 \end{bmatrix} \tag{6}$$

$$f = \frac{f_x + f_y}{2} \tag{7}$$

$$(dX_c)^2 \approx \frac{Z_c^2}{f^2} d\sigma_c^2 \tag{8}$$

$$(dY_c)^2 \approx \frac{Z_c^2}{f^2} d\sigma_c^2 \tag{9}$$

$$(dZ_c)^2 \approx \frac{Z_c^4}{f^2 R^2} d\sigma_r^2 \tag{10}$$

$$\Sigma_C = \begin{bmatrix} dX_c^2 & 0 & 0 \\ 0 & dY_c^2 & 0 \\ 0 & 0 & dZ_c^2 \end{bmatrix} \tag{11}$$

$$R = R_{M\to C}\Sigma_C R_{M\to C}^T \tag{12}$$

The state estimation loop operates recursively in two phases: prediction and measurement update. In the prediction phase, the system projects the *a posteriori* state estimate ($\hat{x}^+$) and error covariance (P^+) from the previous time step ($k-1$) forward to obtain the *a priori* estimates ($\hat{x}^-$, P^-) for the current time step (k):

Predict Step:

$$\hat{x}_k^- = F\hat{x}_{k-1}^+ \tag{13}$$

$$P_k^- = FP_{k-1}^+F^T + Q \tag{14}$$

Upon successful target acquisition in the image plane, the raw spatial measurement z_k is fused with the *a priori* state to compute the refined *a posteriori* state estimate. The Kalman gain (K_k) dynamically weighs the measurement against the state prediction using the adaptive measurement covariance matrix (R):

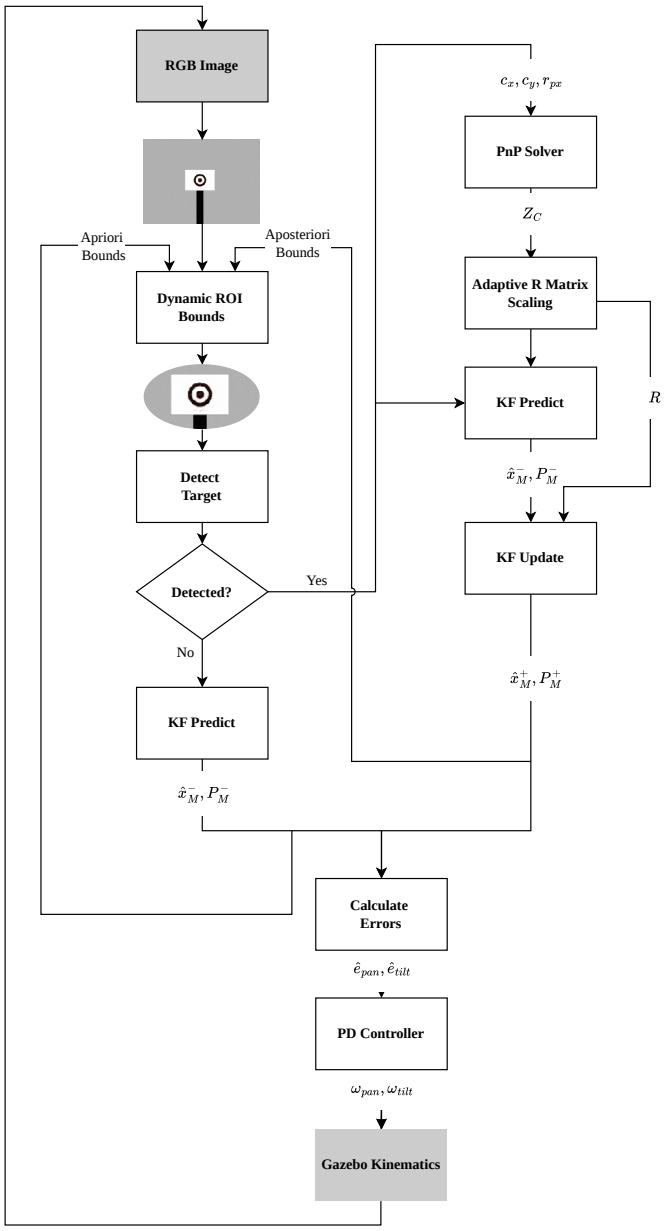


Fig. 2. End-to-end system architecture illustrating the closed-loop coupling between visual feature extraction, dynamic ROI generation, state estimation, and pan-tilt actuation.

Update Step:

$$K_k = P_k^- H^T \left(H P_k^- H^T + R\right)^{-1} \tag{15}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k \left(z_k - H\hat{x}_k^-\right) \tag{16}$$

$$P_k^+ = \left(I - K_k H\right) P_k^- \tag{17}$$

B. Planar Object Pose Estimation with PnP

Inherently, camera frames output 2D sensor output in pixels; to determine the position of a 3D object using the camera frames, either more than one camera has to be used or apriori physical information about the object must be provided. The latter is valid for our case, the target is known to have a shape of a circle with a radius of 9 centimeters.

To obtain a target center pose in the camera frame, we formulate a Non-linear Least Squares problem [17] and solve it using the Levenberg-Marquardt optimization algorithm [18], [19]. For our system, upper, lower, right and left corner pixels of the detected target are represented in both pixel coordinates and target frame coordinates. The optimization algorithm yields a $t_{C\rightarrow T}$ translation vector that's fed to the Kalman Filter for measurement update.

Equations for Homography matrix construction and Levenberg-Marquardt optimization are provided below.

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K\begin{bmatrix} r_1 & r_2 & r_3 & t \end{bmatrix}\begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} \tag{18}$$

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K\begin{bmatrix} r_1 & r_2 & t \end{bmatrix}\begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \tag{19}$$

$$s\mathbf{x} = H\mathbf{X} \tag{20}$$

$$\mathbf{e}(\mathbf{h}_k) = \sum_{i=1}^{n} \left\|\mathbf{x}_i - \hat{\mathbf{x}}_i(\mathbf{h}_k)\right\|^2 \tag{21}$$

$$\mathbf{h}_{k+1} = \mathbf{h}_k - \left(J^T J + \lambda \mathrm{diag}(J^T J)\right)^{-1} J^T \mathbf{e}(\mathbf{h}_k) \tag{22}$$

C. Target Acquisition

To initialize the state estimation filter robustly, the framework executes a spatial voting algorithm across a 20-frame initialization window. For each incoming image, the perception pipeline isolates potential target candidates using HSV color thresholding and a Hough Circle Transform, aggressively filtering out geometries with a fill ratio below 0.2. To systematically reject transient background artifacts, the remaining candidates are grouped using Complete Linkage Clustering [20]. The cluster containing the highest number of members is selected as the true target. This step enables the system to further eliminate outliers and prospective false-positive measurements since clustering objects can discriminate prospective true-positive detections from outliers as candidate detections that've passed the appearance-based visual filters tend to lie in the same cluster [21]. Lastly, its mean spatial center and maximum bounding radius are mapped to a discrete 20x20 2D tracking grid, incrementing that specific cell's total vote count.

After processing the full sequence of 20 frames, the algorithm isolates the grid cell that accumulated the highest vote count, which represents the most spatially and temporally consistent detection region. The initial target state—comprising the starting spatial center coordinates and bounding radius—is then computed by averaging the historical measurements recorded exclusively within this winning cell. This noise-immune, cold-start initialization guarantees that the Kalman filter is seeded with a highly reliable spatial measurement before the system transitions into the continuous tracking phase.

D. Feature-driven Object Detection & Dynamic ROI

State-of-the-art deep learning detectors (e.g., YOLO, MobileNet) excel in modern image processing tasks [1], [2],

[22], but model performance is highly degraded in accurate detection of small and fast-moving objects since visual features of the targets are comprimised in fast movements due to motion-induced blur and the fact that small objects rarely provide distinct geometric features that deep learning-based models heavily rely on [6].

In this article, we implemented a classic computer vision alternative that utilizes visual features preserved under chaotic motion for accurate and robust target detection. Our pipeline leverages the color, size and shape of the target for detection and incorporate complete linkage clustering [20], [21] to eliminate outliers, finally the detection closest to the pixel projection of the latest estimated target state is selected as the detection.

Color information is filtered with an HSV filter tuned for the target color scheme and a fill ratio threshold filter. A Hough transform filter is utilized to detect prospective targets, the maximum radius in pixels is determined by using the apriori knowledge about the target and depth estimate obtained by the latest camera measurement. The fact that target radius is 9 cm is known apriori, using the intrinsic parameters of the pinhole camera model [17] we can create an upper bound in pixels for the target detection described in equation 23.

$$r_{\text{px}} = \max\left(\frac{f_x r_{\text{cm}}}{Z}, \frac{f_y r_{\text{cm}}}{Z}\right) + r_{\text{offset}} \tag{23}$$

The offset parameter in equation 23 is added to account for depth measurement errors in the camera measurements and is tuned practically.

E. State-predicted Dynamic Region of Interest(ROI) Generation

For computationally effective object detection immune to false-positive detections, our system leverages target pose estimates of the Kalman Filter to create a dynamic search region for prospective detections. To construct a search region, Kalman Filter position estimate mean $\mu_{M\rightarrow T}$ and error covariance $\Sigma_{M,3D}$ are projected onto the image plane using 1st covariance propagation described in the equations 24-25.

Linear covariance propogation of an ellipsoid onto 2-D plane outputs an ellipse characterized by eigen values, λ_u, λ_v which signify semi-major and semi-minor axis length of the ellipse, and eigen vectors e_u, e_v which signify semi-major and semi-minor axis directions. α_{ROI} value described in these equations is a tunable parameter determining the search region area.

Kalman Filter-based ROI determination is frequently used in real-time tracking tasks where the tracking algorithm starts with an initial ROI and the ROI is updated based on Kalman Filter state and uncertainty estimations [7]. ROI filtering leads to faster computation and enables the system to utilize more computationally heavy methods and is utilized in current articles in realtime vision-based tracking [23].

$$p^C = R_{C\rightarrow M} p^M + t_C \tag{24}$$

$$J_C = \begin{bmatrix} \frac{\partial u}{\partial p_x} & \frac{\partial u}{\partial p_y} & \frac{\partial u}{\partial p_z} \\ \frac{\partial v}{\partial p_x} & \frac{\partial v}{\partial p_y} & \frac{\partial v}{\partial p_z} \end{bmatrix} = \begin{bmatrix} \frac{f_x}{p_z} & 0 & -\frac{f_x p_x}{p_z^2} \\ 0 & \frac{f_y}{p_z} & -\frac{f_y p_y}{p_z^2} \end{bmatrix} \tag{25}$$

$$\mu_{u,v} = \begin{bmatrix} \mu_u \\ \mu_v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{p_x}{p_z} \\ \frac{p_y}{p_z} \\ 1 \end{bmatrix} \tag{26}$$

$$\Sigma_{2D} = J_C R_{C\rightarrow M} \Sigma_{M,3D} R_{C\rightarrow M}^T J_C^T \tag{27}$$

$$l_u = \alpha_{roi}\sqrt{\lambda_u} \tag{28}$$

$$l_v = \alpha_{roi}\sqrt{\lambda_v} \tag{29}$$

F. Target Tracking & Closed-loop Control

In the proposed system, laser pointer axis is aligned with the z-axis of the laser, to obtain the target center position in the laser frame, calculate angular errors, and generate angular velocity commands demonstrated in the system block diagram, following equations are utilized.

$$p^L = T_{M\rightarrow L} p^M \tag{30}$$

$$e_{pan} = tan^{-1}(\frac{p_y^L}{p_z^L}) \tag{31}$$

$$e_{tilt} = tan^{-1}(\frac{p_x^L}{p_z^L}) \tag{32}$$

$$\omega_{pan,k} = K_p e_{pan,k} + K_d(\frac{e_{pan,k} - e_{pan,k-1}}{\Delta t}) \tag{33}$$

$$\omega_{tilt,k} = K_p e_{tilt,k} + K_d(\frac{e_{tilt,k} - e_{tilt,k-1}}{\Delta t}) \tag{34}$$

G. Self-evaluation Metric

In this research, we focused on obtaining robust self-evaluation metrics that utilize scenario constraints and physical information provided. Normally, precision percentage calculation includes knowledge of ground truth target positions at each evaluation interval and an error threshold that indicates whether the laser points on the target.

For scenarios where ground truth measurements aren't available, we propose a novel evaluation strategy for precision percentage calculations. This new error metric returns a pixel distance value threshold that when constrained with simulation assumptions, guarantees that the laser points on the target without ground truth measurements.

Our metric determines camera pixel frame coordinates of 2 points and calculates the pixel distance between these two points. First, it takes the pixel coordinates of the latest target detection. Second, it calculates the line on the pixel frame that corresponds to the possible laser pointer positions given the max depth uncertainty of 100 cm. Another important point is that in our simulations, laser and camera frames are parallel, laser frame is offsetted by a y_{offset} from the camera frame. Offset in laser v pixels given the depth uncertainty is calculated using the equation 35

$$e_{l,max} = \frac{f_v y_{offset} \times 100}{Z_c^2 - 100 Z_c} \tag{35}$$

For the given depth uncertainty of 100cm, smallest possible target radius in px can be calculated using equations 36 - 37.

$$f_{min} = min(f_x, f_y) \tag{36}$$

$$r_{px,min} = \frac{f_{min} R}{Z_c + 100} \tag{37}$$

Laser point correspondence on the pixel coordinates calculated by the estimated depth and its distance to target pixel center can be calculated using equations 38 - 39

$$(u_l, v_l) = (c_x, c_y + \frac{f_v R}{Z_c}) \tag{38}$$

$$d_{px} = \sqrt{(u_l - u_t)^2 + (v_l - v_t)^2} \tag{39}$$

Then we can propose a radius threshold in pixels that when satisfied, guarantees that the target laser pointer is on the target, which is found using the equation 40, which points if the calculated radius is smaller than the difference between smallest possible radius and maximum laser pointer coordinate deviation, than the laser is guaranteed to point on target.

$$r_{th} = \frac{f_{min} R}{Z_c + 100} - \frac{f_v y_{offset} \times 100}{Z_C^2 - 100 Z_c} \tag{40}$$

In the experiments, we used this metric to calculate precision. When mathematical assumptions are satisfied, which we think is much easier to verify or determine than the daunting ground truth measurement process, constitutes a highly accurate self-evaluated precision metric for model performance evaluation.

III. Experiments and Results

To rigorously evaluate the proposed tracking architecture, Monte Carlo simulations were conducted across multiple dynamic scenarios within the Gazebo simulation environment. To explicitly mirror real-world deployment constraints where external motion capture or ground-truth data is unavailable, the system's performance is quantified using a mathematically derived, sensor-only evaluation approach. Detailed parameters of the simulation environment and the derivation of these experimental metrics are described in the following subsections.

A. Experimental Setup

1) Physics Engine: Monte Carlo simulations were conducted within the Gazebo simulation environment to rigorously validate system performance. Gazebo was selected for its robust physics engine and high-fidelity sensor plugins, enabling the accurate replication of real-world conditions, such as lighting, mechanical vibration, and friction.

2) Software Framework: The proposed tracking architecture is implemented within the ROS 2 framework for the compatibility of the simulation environment and the applicability in real-time. Specifically, ROS2 action servers are utilized for state management in different tracking scenarios.

3) Cluttered Simulation Environment: To rigorously evaluate the robustness of the visual tracking pipeline against false-positive detections, visually similar distractor objects (e.g., black cones and traffic signs) were intentionally introduced into the field of view throughout the simulation. This highly cluttered setting explicitly stress-tests the performance of the Kalman filter's state-predicted dynamic ROI generation, as well as the outlier rejection capabilities of the object detection pipeline.

B. Scenarios

1) Linear-Forward: In the linear-forward scenario, the target is initialized at a depth of 7 to 8 meters from the camera. Following initial target acquisition, the object translates linearly along the depth axis directly toward the tracking platform. To rigorously evaluate the system's operational stability, this scenario was executed across 30 independent Monte Carlo trials. For each trial, the initial starting depth was uniformly sampled between 7 and 8 meters, with a constant approach velocity randomly assigned between 0.3 and 0.5 m/s. This stochastic configuration explicitly stress-tests the scale-adaptive limits of the dynamic ROI generation and the transient response of the Kalman filter under varying continuous approach speeds.

2) Drive-by: In the lateral drive-by scenario, the target is initialized at a depth uniformly sampled between 7 and 8 meters. Unlike the linear-forward case, the target does not directly face the camera; its initial declination angle is uniformly selected between $15°$ and $30°$. Following initial acquisition, the object traverses the tracking platform's field of view at a constant velocity randomly assigned between 0.3 and 0.5 m/s. This scenario was similarly evaluated across 30 independent Monte Carlo trials to stress-test the system's tracking persistence, state-predicted dynamic ROI generation, and adaptive measurement covariance propagation under conditions characterized by reduced line-of-sight and rapid spatial profile changes.

3) Sinusoidal: In the sinusoidal trajectory scenario, the tracking platform initially faces the target, with the target's starting position subjected to an initial lateral offset uniformly sampled between -1.0 and 1.0 m. Upon target acquisition, the object follows a continuous non-linear trajectory with a constant linear velocity randomly assigned between 0.3 and 0.5 m/s, while its angular velocity is dynamically governed by a harmonic profile:

$$\omega(t) = 1.25 \cdot \sin(t)$$

where t denotes the elapsed simulation time in seconds. Evaluated across 30 independent Monte Carlo trials, this highly dynamic configuration explicitly stress-tests the transient performance of the Extended Kalman Filter, the continuous-to-discrete process noise (Q) model, and the stability of the dynamic ROI generation loop under chaotic, non-linear target perturbations.

C. Results

1) Global Perception Pipeline & Detection Benchmark: Across 180 Monte Carlo trials (comprising two measurement

covariance configurations, three trajectory scenarios, and 30 trials per scenario), the object detection module consistently achieved $> 99\%$ detection accuracy with zero false positives. This operational robustness is enabled by a feature-driven detection pipeline optimized for rapid execution under high-speed, chaotic motions of small targets, paired with state-predicted dynamic ROI generation that isolates prospective measurement regions.

Having established that visual target acquisition is virtually lossless, the remainder of this section focuses strictly on state estimation precision—specifically evaluating image plane coordinate errors (u, v) and total 2D scalar pixel distance errors against the mathematically derived worst-case bound (r_{th}).

2) Linear-forward Scenario Analysis: Target tracking results for the linear-forward scenario are visually demonstrated in Fig. 3 and summarized in Table I. A comparative analysis

TABLE I
LINEAR-FORWARD SCENARIO ERROR METRICS ($N = 30$ TRIALS)

Metric	Adaptive-R	Static-R
u_{err} (px)	2.23 ± 0.84	1.88 ± 0.56
v_{err} (px)	2.67 ± 0.14	2.90 ± 0.21
pixel_dist$_{\text{err}}$ (px)	3.54 ± 0.48	3.49 ± 0.29
Precision (%)	$100.00\% \pm 0.00\%$	$99.85\% \pm 0.24\%$

between the Adaptive-R and Static-R configurations demonstrates exceptional tracking performance across both models, maintaining low overall pixel distance errors (3.54 ± 0.48 px and 3.49 ± 0.29 px, respectively). The global perception pipeline consistently isolates the target within the dynamic region of interest (ROI) by leveraging appearance-based features robust to high-speed translational motion. Furthermore, the PD control loop smoothly regulates platform pan-tilt actuation under varying target velocities.

Notably, while Static-R yields a slightly lower horizontal mean error ($u_{\text{err}} = 1.88$ px), the Adaptive-R configuration achieves superior vertical accuracy ($v_{\text{err}} = 2.67$ px) and guarantees a 100.00% precision rate strictly within the worst-case error bound (r_{th})

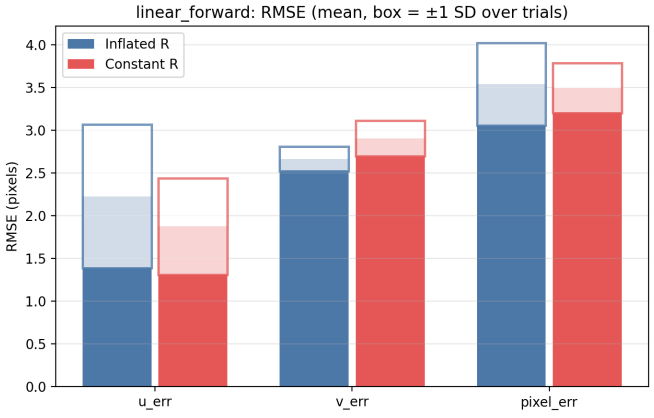


Fig. 3. Comparative RMSE metrics for the linear-forward scenario across $N = 30$ Monte Carlo trials. Solid bars indicate mean RMSE values, while outer boxes represent ± 1 standard deviation bounds.

3) Lateral Drive-by Scenario Error Metrics ($N = 30$ Trials): As illustrated in Fig. 4 and summarized in Table II, in the lateral drive-by scenario, rapid cross-frame target traversal elevates horizontal pixel tracking errors (u_{err}) across both configurations, driving total 2D pixel distance errors to 8.31 ± 1.34 px for Adaptive-R and 7.55 ± 1.02 px for Static-R.

Despite initial declination angles ($15°$–$30°$) and reduced line-of-sight exposure, both models maintain exceptional target lock, with Static-R and Adaptive-R achieving 99.70% and 98.95% precision rates strictly within the worst-case bound (r_{th}), respectively. Notably, vertical tracking accuracy remains highly stable across both configurations ($v_{\text{err}} \approx 2.51$–$2.52$ px), confirming that high lateral motion predominantly impacts horizontal image-plane error dynamics while the state-predicted dynamic ROI successfully prevents target loss.

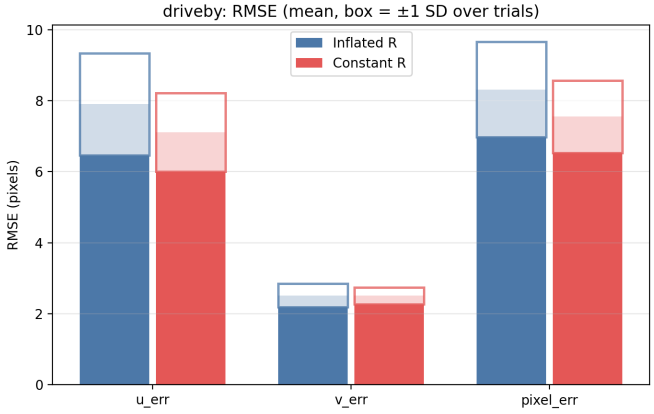


Fig. 4. Comparative RMSE metrics for the lateral drive-by scenario across $N = 30$ Monte Carlo trials. Solid bars indicate mean RMSE values, while outer boxes represent ± 1 standard deviation bounds.

TABLE II
LATERAL DRIVE-BY SCENARIO ERROR METRICS ($N = 30$ TRIALS)

Metric	Adaptive-R	Static-R
u_{err} (px)	7.90 ± 1.44	7.11 ± 1.11
v_{err} (px)	2.52 ± 0.33	2.51 ± 0.24
pixel_dist$_{\text{err}}$ (px)	8.31 ± 1.34	7.55 ± 1.02
Precision (%)	$98.95\% \pm 1.75\%$	$99.70\% \pm 0.67\%$

4) Sinusoidal Scenario Error Metrics ($N = 30$ Trials): The sinusoidal motion profile represents the most challenging operational benchmark, designed to mirror real-world agile tracking environments characterized by continuous non-linear acceleration and harmonic lateral maneuvers ($\omega(t) = 1.25 \cdot \sin(t)$). Under these highly dynamic conditions, the Adaptive-R configuration significantly outperforms the Static-R model in both tracking accuracy and overall filter stability. Tracking architecture performance is illustrated in the Fig. 5 and summarized numerically in Table III.

Specifically, the Adaptive-R scaling mechanism limits horizontal error variance to 0.84 px ($u_{\text{err}} = 9.59 \pm 0.84$ px) and overall scalar pixel distance variance to 0.82 px (10.08 ± 0.82 px). Conversely, the Static-R filter suffers from severe error variance inflation ($u_{\text{err}} = 11.36 \pm 3.61$ px and total pixel error $= 11.73 \pm 3.55$ px), demonstrating pronounced phase lag

TABLE III
SINUSOIDAL TRAJECTORY SCENARIO ERROR METRICS ($N = 30$ TRIALS)

Metric	Adaptive-R	Static-R
u_{err} (px)	9.59 ± 0.84	11.36 ± 3.61
v_{err} (px)	3.07 ± 0.30	2.87 ± 0.19
pixel_dist$_{\text{err}}$ (px)	10.08 ± 0.82	11.73 ± 3.55
Precision (%)	$89.94\% \pm 3.60\%$	$84.63\% \pm 9.84\%$

and state drift during rapid directional swings. Consequently, Adaptive-R maintains a superior bounded precision rate of $89.94 \pm 3.60\%$, compared to $84.63 \pm 9.84\%$ for Static-R. This notable performance gap demonstrates that dynamically inflating measurement noise as a function of depth uncertainty ($e_z \propto Z^2$) prevents filter divergence under aggressive, real-world target maneuvers.

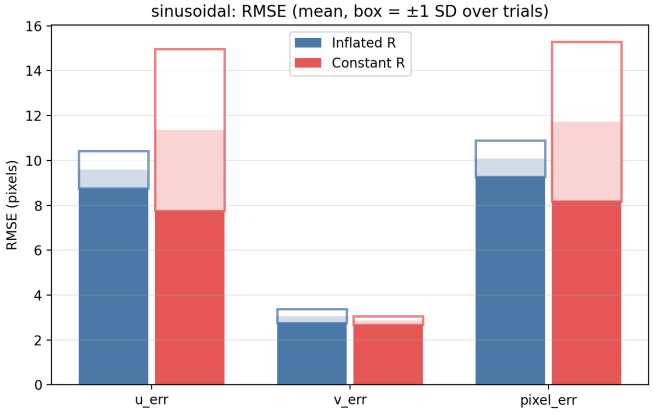


Fig. 5. Comparative RMSE metrics for the sinusoidal scenario across $N = 30$ Monte Carlo trials. Solid bars indicate mean RMSE values, while outer boxes represent ± 1 standard deviation bounds.

5) Comparative Variance Analysis (Static-R vs. Adaptive-R): To evaluate the operational limits of the proposed visual state estimator, we contrast the performance of the Adaptive-R matrix scaling mechanism against a conventional Static-R matrix baseline across all 180 Monte Carlo trials ($N = 30$ per scenario configuration).

While both configurations demonstrate comparable baseline stability in simplistic trajectory profiles, the Adaptive-R formulation exhibits vastly superior robustness, variance suppression, and error-bounding guarantees as target kinematics transition to non-linear, high-speed regimes representative of real-world tracking environments.

The critical performance divergence between the two state estimators manifests under continuous non-linear acceleration ($\omega(t) = 1.25 \cdot \sin(t)$)—the scenario closest to real-world agile tracking applications. Under these conditions, the Static-R filter suffers from severe phase lag and state drift, whereas the Adaptive-R mechanism dynamically inflates measurement noise as depth-dependent uncertainty expands ($e_z \propto Z^2$), preventing filter divergence. Quantitative comparisons in the sinusoidal scenario highlight the substantial gains in both tracking accuracy and structural variance suppression.

These empirical results confirm that while a static measurement covariance matrix (R) can suffice under unperturbed linear conditions, it introduces unacceptable state jitter at far depths and severe phase lag during non-linear target maneuvers. By dynamically tuning R based on real-time depth estimation uncertainty, the Adaptive-R architecture guarantees filter convergence, minimizes tracking variance, and maintains tight precision bounds under aggressive target kinematics.

IV. CONCLUSION AND FUTURE WORK

A. Conclusion

In this work, we presented a resource-efficient, low-latency vision-based target tracking framework that combines classical computer vision features with adaptive state estimation. While state-of-the-art deep learning models (e.g., lightweight convolutional networks and single-shot YOLO detectors) offer automated feature extraction, their computational overhead compromises control loop stability during rapid, highly dynamic maneuvers. To address these constraints, our perception stack leverages a structured grid-based acquisition scheme, appearance-based features resilient to fast motion blur, complete linkage clustering for outlier rejection, and state-predicted dynamic region of interest (ROI) generation. This integrated setup effectively eliminates false-positive visual artifacts and ensures continuous target lock without relying on high-latency deep learning architectures.

To improve tracking precision under aggressive target dynamics, the framework incorporates a physics-informed Adaptive-R covariance matrix scaling mechanism. By applying a first-order Taylor expansion to map image-plane spatial dimensions to 3D pose uncertainty, the measurement noise covariance dynamically scales as a quadratic function of depth uncertainty ($e_z \propto Z^2$). Coupled with a tuned Proportional-Derivative (PD) controller regulating pan-tilt actuator commands, the adaptive architecture minimizes state phase lag and prevents filter divergence.

Furthermore, to address the challenge of validating tracking performance in environments where external motion capture or ground-truth coordinates are unavailable, we introduced a novel, sensor-only self-evaluation metric. By deriving a mathematically conservative worst-case pixel error bound (r_{th}) based on the physical limits of the camera optics, target geometry, and depth estimation error, the proposed methodology enables rigorous performance verification without daunting and error-prone ground-truth labeling. Monte Carlo evaluations across 180 simulation trials demonstrated that under complex, non-linear sinusoidal trajectories, the Adaptive-R formulation achieves a $4.3\times$ suppression in pixel error variance (± 0.82 px vs. ± 3.55 px) and maintains an 89.94% bounded precision rate within strict r_{th} limits compared to a static covariance baseline.

B. Future Work

Future research will focus on extending the capabilities and real-world applicability of the proposed tracking architecture along three main directions:

1) **Loss of Line-of-Sight Stress-Testing:** Evaluating state estimator resilience under prolonged target occlusions

and complete loss of line-of-sight scenarios by propagating covariance ellipses without measurement updates.

2) **Real-World Self-Evaluation Correlation:** Conducting hardware-in-the-loop experiments with external motion capture (e.g., OptiTrack/Vicon) to analyze the exact statistical correlation between our derived sensor-only precision metric (r_{th}) and true physical ground-truth measurements.
3) **Comparative Edge Deep Learning Benchmark:** Benchmarking the computational throughput, latency, and tracking precision of our classical feature-driven pipeline against modern edge-optimized deep learning detectors (such as YOLOv8-Nano and MobileNetV4) on embedded robotic hardware.

ACKNOWLEDGMENT

The authors would like to thank...

REFERENCES

[1] Z.-Q. Zhao, P. Zheng, S.-t. Xu, and X. Wu, "Object detection with deep learning: A review," *IEEE transactions on neural networks and learning systems*, vol. 30, no. 11, pp. 3212–3232, 2019.

[2] J. Redmon and A. Farhadi, "Yolov3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[3] L. Zhou, Q. Li, G. Huo, and Y. Zhou, "Image classification using biomimetic pattern recognition with convolutional neural networks features," *Computational intelligence and neuroscience*, vol. 2017, no. 1, p. 3792805, 2017.

[4] P. Mittal, "A comprehensive survey of deep learning-based lightweight object detection models for edge devices: P. mittal," *Artificial Intelligence Review*, vol. 57, no. 9, p. 242, 2024.

[5] P. R. Singh, R. Gottumukkala, A. S. Maida, A. B. Barhorst, and V. Gopu, "Physics-guided fusion for robust 3d tracking of fast moving small objects," 2025. [Online]. Available: https://arxiv.org/abs/2510.20126

[6] Z. Zhang, F. Wu, Y. Qiu, J. Liang, and S. Li, "Tracking small and fast moving objects: A benchmark," in *Computer Vision – ACCV 2022*, L. Wang, J. Gall, T.-J. Chin, I. Sato, and R. Chellappa, Eds. Cham: Springer Nature Switzerland, 2023, pp. 552–569.

[7] C. F. Graetzel, B. J. Nelson, and S. N. Fry, "A dynamic region-of-interest vision tracking system applied to the real-time wing kinematic analysis of tethered drosophila," *IEEE Transactions on Automation Science and Engineering*, vol. 7, no. 3, pp. 463–473, 2009.

[8] O. Korkalo and T. Takala, "Measurement noise model for depth camera-based people tracking," *Sensors*, vol. 21, no. 13, p. 4488, 2021.

[9] J. Li, X. Xu, Z. Jiang, and B. Jiang, "Adaptive kalman filter for real-time visual object tracking based on autocovariance least square estimation," *Applied Sciences*, vol. 14, no. 3, p. 1045, 2024.

[10] C. Spampinato, S. Palazzo, and D. Giordano, "Evaluation of tracking algorithm performance without ground-truth data," in *2012 19th IEEE international conference on image processing*. IEEE, 2012, pp. 1345–1348.

[11] C. E. Erdem, B. Sankur, and A. M. Tekalp, "Performance measures for video object segmentation and tracking," in *Visual Communications and Image Processing 2003*, T. Ebrahimi and T. Sikora, Eds., vol. 5150, International Society for Optics and Photonics. SPIE, 2003, pp. 29 – 40. [Online]. Available: https://doi.org/10.1117/12.509859

[12] R. E. Kalman, "A new approach to linear filtering and prediction problems," 1960.

[13] G. Welch, G. Bishop *et al.*, "An introduction to the kalman filter," 1995.

[14] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with applications to tracking and navigation: theory algorithms and software*. John Wiley & Sons, 2001.

[15] X. R. Li and V. P. Jilkov, "Survey of maneuvering target tracking. part i. dynamic models," *IEEE Transactions on aerospace and electronic systems*, vol. 39, no. 4, pp. 1333–1364, 2003.

[16] T. Collins and A. Bartoli, "Infinitesimal plane-based pose estimation," *International journal of computer vision*, vol. 109, no. 3, pp. 252–286, 2014.

[17] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on pattern analysis and machine intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.

[18] K. Madsen, H. Nielsen, O. Tingleff, and M. MODELLING, "Imm methods for non-linear least squares problems," *Technical University of Denmark, DTU*, 2004.

[19] E. Eade, "Gauss-newton/levenberg-marquardt optimization," *Tech. Rep.*, 2013.

[20] S. C. Johnson, "Hierarchical clustering schemes," *Psychometrika*, vol. 32, no. 3, p. 241–254, 1967.

[21] J. A. Lara, D. Lizcano, V. Rampérez, and J. Soriano, "A method for outlier detection based on cluster analysis and visual expert criteria," *Expert Systems*, vol. 37, no. 5, p. e12473, 2020.

[22] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[23] M. Abdelkader, K. Gabr, I. Jarraya, A. AlMusalami, and A. Koubaa, "Smart-track: a novel kalman filter-guided sensor fusion for robust uav object tracking in dynamic environments," *IEEE Sensors Journal*, vol. 25, no. 2, pp. 3086–3097, 2024.